Faculty of Engineering & Technology

Electrical & Computer Engineering Department

Machine Learning- ENCS4380

Assignment #1

Prepared by: Laith iyad, Saleh Akram

ID Number: 1221123 ,1220641

Instructor: Dr. Yazan Abu-farha

Section: 1,2

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Introduction:

In this assignment, we worked with a made-up (synthetic) customer dataset that includes information like age, gender, income, how long the customer has been with the company, the type of product they use, and whether they are likely to leave (churn) or stay.

we used Python to complete the tasks, along with libraries such as pandas for handling the data, seaborn, and matplotlib for making charts and visualizations. These tools helped me explore the data, understand patterns, and find insights about customer behaviour and churn.

The results and explanations for each step are shown below.

Results of the Steps for the Assignment:

1. Data Loading and Initial Inspection:

```
   CustomerID   Age  Gender    Income  Tenure  ProductType  SupportCalls  ChurnStatus
0   CUST0000  59.0       0  151203.0     4.0            0           1.0            0
1   CUST0001  69.0       0   58332.0     6.0            1           9.0            0
2   CUST0002  46.0       1  149481.0     2.0            0          12.0            0
3   CUST0003  32.0       1  115937.0     1.0            1          13.0            0
4   CUST0004  60.0       0  103929.0     4.0            1           5.0            0
```

The figure above shown the first 5 rows in the data frame in the pandas data frame that load from the csv file. [df.head()]

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   CustomerID    3500 non-null   object
 1   Age           3325 non-null   float64
 2   Gender        3500 non-null   int64
 3   Income        3328 non-null   float64
 4   Tenure        3325 non-null   float64
 5   ProductType   3500 non-null   int64
 6   SupportCalls  3329 non-null   float64
 7   ChurnStatus   3500 non-null   int64
dtypes: float64(4), int64(3), object(1)
memory usage: 218.9+ KB
None
```

In this figure, the function info() show us the information about data frame, columns, non-empty values and the data type of each columns.

```
                Age       Gender        Income       Tenure  ProductType  SupportCalls  ChurnStatus
count  3325.000000  3500.000000  3.328000e+03  3325.000000  3500.000000   3329.000000  3500.000000
mean     43.606617     0.495714  1.406860e+05     5.041504     0.298857     10.078702     0.044857
std      14.931487     0.500053  4.333271e+05     2.571029     0.457822     21.735374     0.207020
min      14.000000     0.000000  2.503700e+04     0.000000     0.000000      1.000000     0.000000
25%      31.000000     0.000000  5.653025e+04     3.000000     0.000000      3.000000     0.000000
50%      43.000000     0.000000  8.953250e+04     5.000000     0.000000      7.000000     0.000000
75%      56.000000     1.000000  1.215025e+05     7.000000     1.000000     11.000000     0.000000
max      69.000000     1.000000  5.004849e+06     9.000000     1.000000    200.000000     1.000000
```

Using describe function display the above information about the columns. We can notes that there are different scales, so each column has different scale from the other columns.

2. Handling Missing Data:

   First, we check the missing data in the table by using .isnull().sum() function, that result the following data.

```
CustomerID       0
Age            175
Gender           0
Income         172
Tenure         175
ProductType      0
SupportCalls   171
ChurnStatus      0
dtype: int64
```

   These data means that there are a missing data in the age, income, tenure and support calls columns, and there are no missing data in Customer ID, Gender, Product Type and Churn Status columns.

   After that, we use isnull().any(axis=1).sun() to display the numbers of the rows that has al least one missing data to know the effect of the missing data in the whole data set, then the total rows were 643 rows that constituted approximately 18% of the dataset.

   Now, we handle missing values by imputation them with the mean value, because the dataset is small, and deleting rows would remove too much data. The mean represents a reasonable estimate.

   After filling missing data, we can see that there are no rows that have least one missing data.

```
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   CustomerID    3500 non-null   object
 1   Age           3500 non-null   float64
 2   Gender        3500 non-null   int64
 3   Income        3500 non-null   float64
 4   Tenure        3500 non-null   float64
 5   ProductType   3500 non-null   int64
 6   SupportCalls  3500 non-null   float64
 7   ChurnStatus   3500 non-null   int64
dtypes: float64(4), int64(3), object(1)
memory usage: 218.9+ KB
None
```

3. Handling Outliers:

In this step we will identify the numerical outliers using Z-Score without modify the scaling, so we used the following equations to find the upper and lower limit for each column, and any feature in specific column exceeded the upper or lower limit will consider it as an outlier.

Equations:

upper = df[col].mean() + 2 * df[col].std()

lower = df[col].mean() - 2 * df[col].std()

```
Outliers detected in Age: 0

Outliers detected in Income: 50

Outliers detected in Tenure: 0

Outliers detected in SupportCalls: 70
```

As illustrated in figures above, there are 50 outliers in income and 70 in support calls columns.

After that we decided to remove each outlier and do not imputation them, to have more accurate data set and to not effect on the data set relations.

4. Feature Scaling:

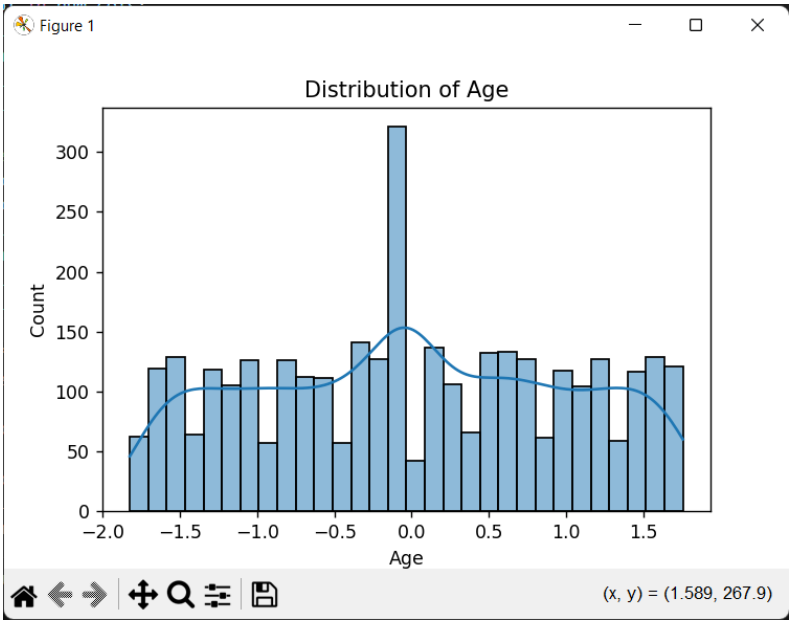
Now, we choose Standardization (Z-score) to scaling the numerical features, and the whole data was near to the mean [range: -2.5 to 2.5].

	CustomerID	Age	Gender	Income	Tenure	ProductType	SupportCalls	ChurnStatus
0	CUST0000	1.066394	0	0.757336	-0.407433	0	-1.482697	0
1	CUST0001	1.754495	0	-0.450067	0.389283	1	0.473398	0
2	CUST0002	0.171862	1	0.734948	-1.204148	0	1.206934	0
3	CUST0003	-0.791479	1	0.298847	-1.602506	1	1.451446	0
4	CUST0004	1.135204	0	0.142733	-0.407433	1	-0.504649	0
...	...	...	...	...	...	...	...	...
3495	CUST3495	-0.240998	1	-0.721954	0.389283	0	1.695957	0
3496	CUST3496	1.479255	0	-0.044434	-1.204148	1	-1.482697	0
3497	CUST3497	-0.309808	1	0.220777	1.185998	0	1.695957	0
3498	CUST3498	-0.653859	1	0.672947	-0.009075	0	1.695957	0
3499	CUST3499	-1.479581	0	0.476062	-0.009075	0	0.717910	0

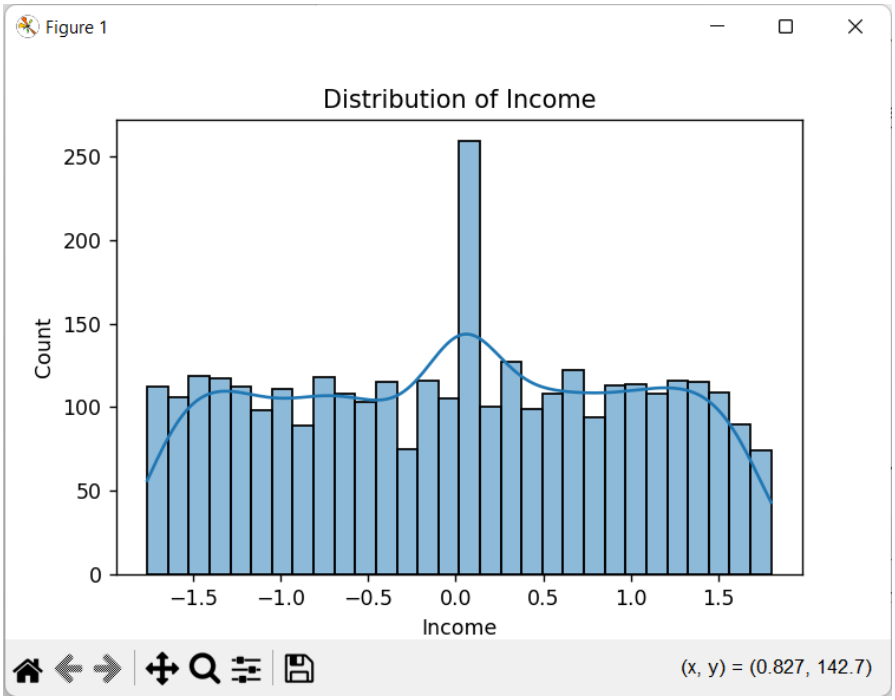
Figure shows the new values after apply the scaling on some numerical feature.

Task5:

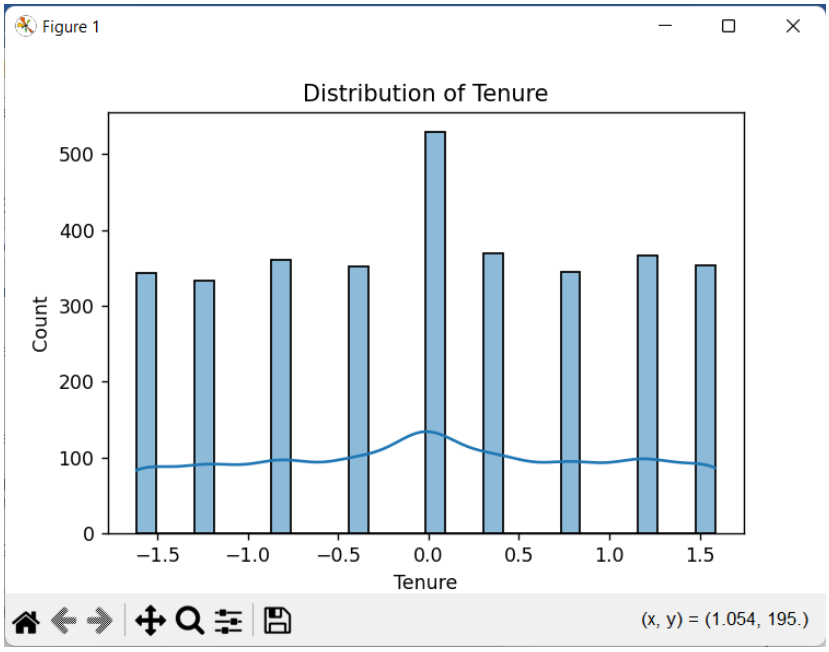
Univariate Analysis:



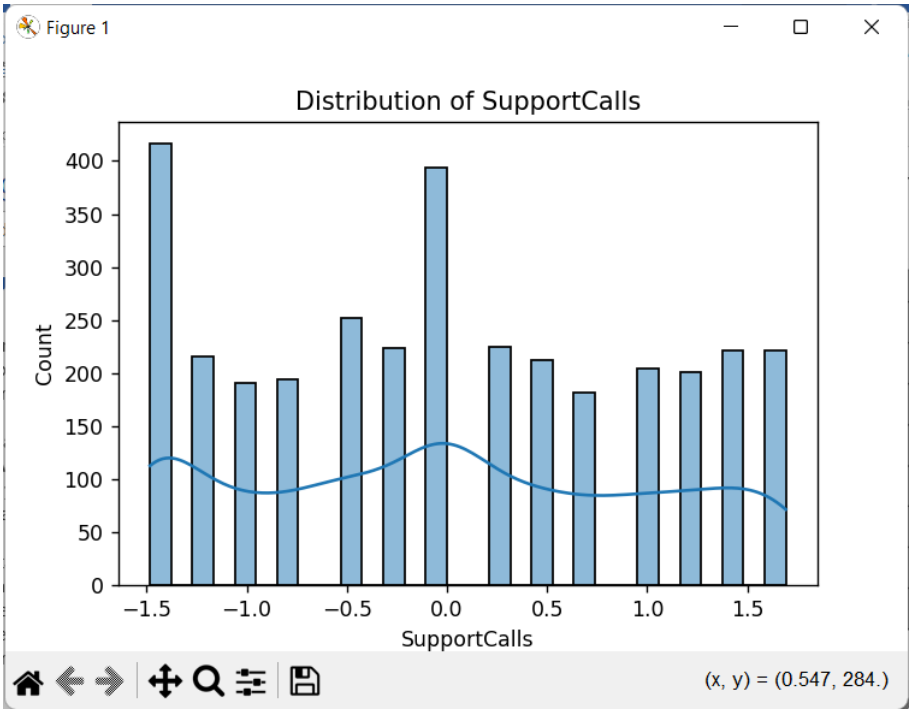
This is a histogram which shows the relationship between the (scaled) age and number category as we show the mean is almost equal 0 and the other value is around it.



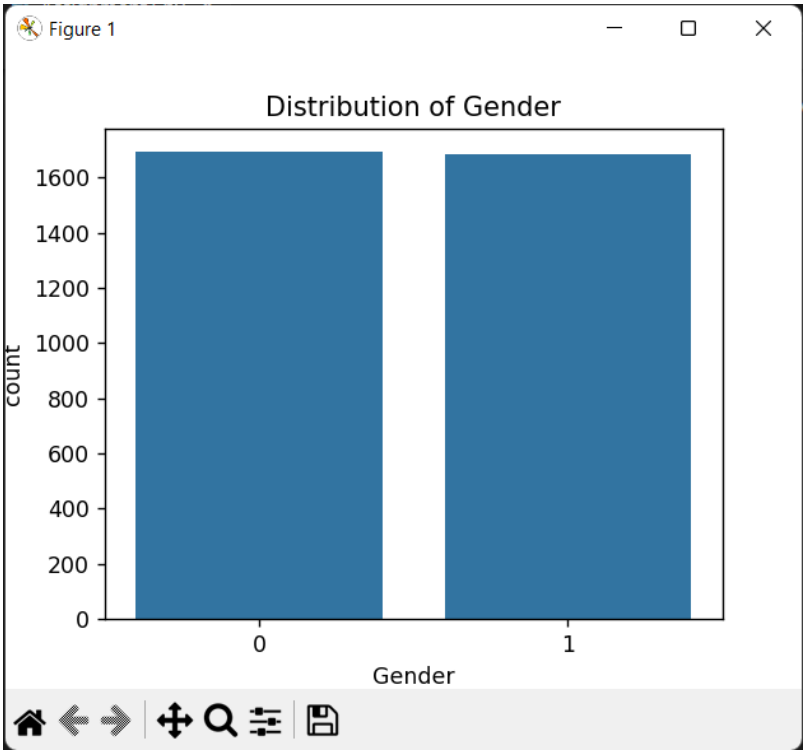
This is a histogram which shows the relationship between the (scaled) Income and number category as we show the mean is almost equal 0 or upper and the other value is around it.



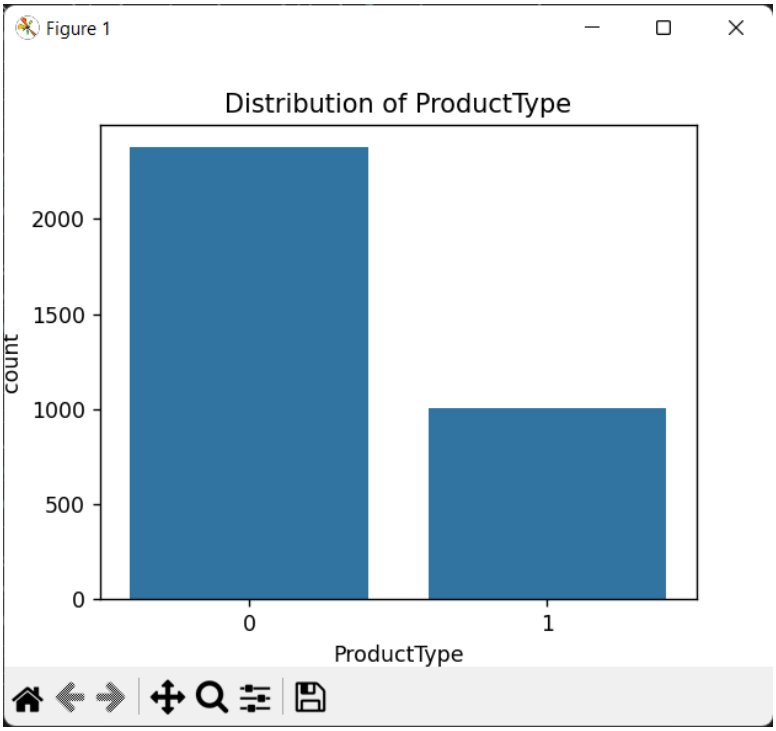
This is a histogram which shows the relationship between the (scaled) Tenure and number category as we show the mean is almost equal 0 and the other value is around it



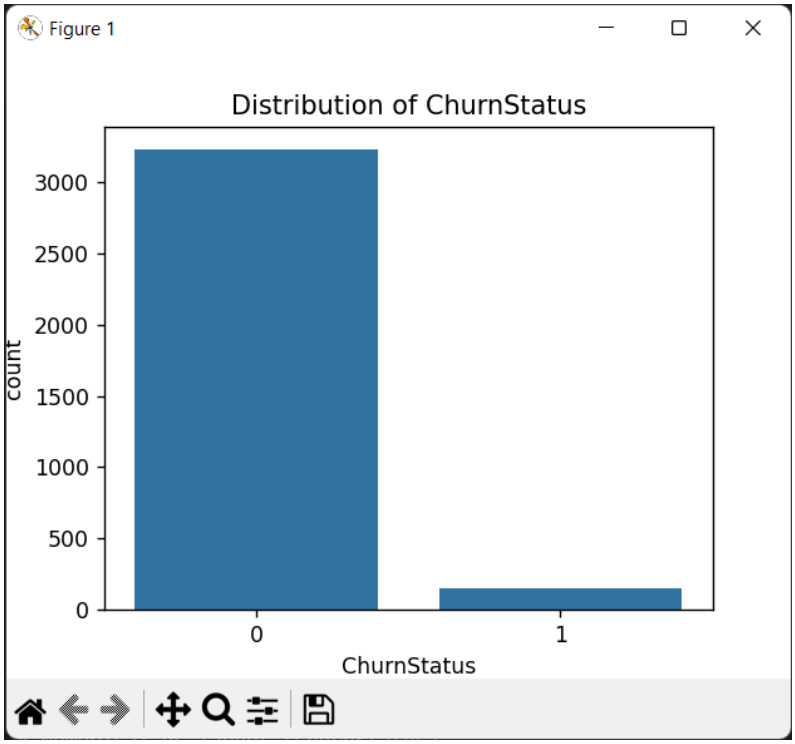
This is a histogram which shows the relationship between the (scaled) SupportCalls and number category as we show the mean is almost equal 0 and the other value is around it. But it is more towards lower values.



This is a histogram which shows the relationship between the (scaled) Gender and number as we show it not effect 50% of the count is male and 50% is female.

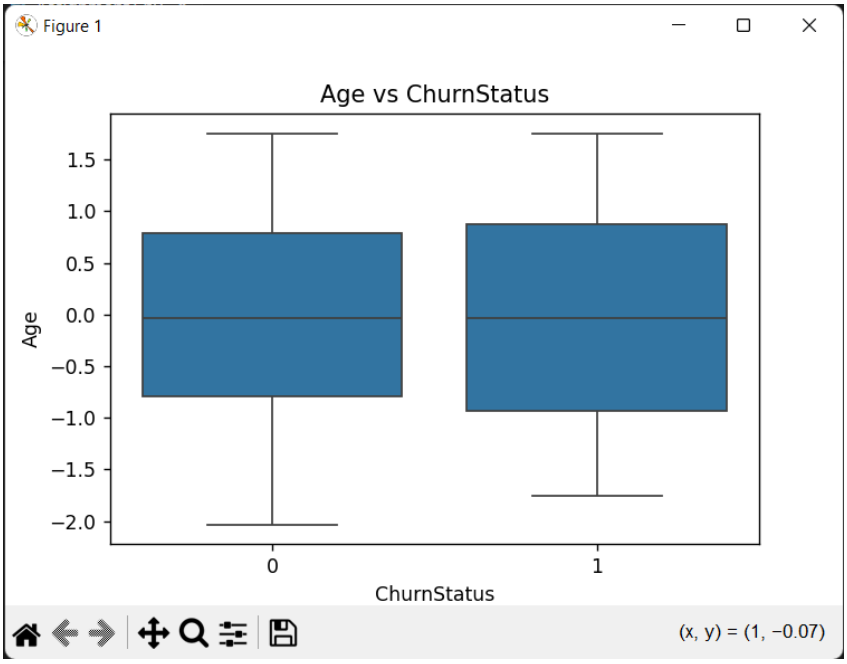


This is a histogram which shows the relationship between the (scaled) productType and number as we show the most number is Basic.

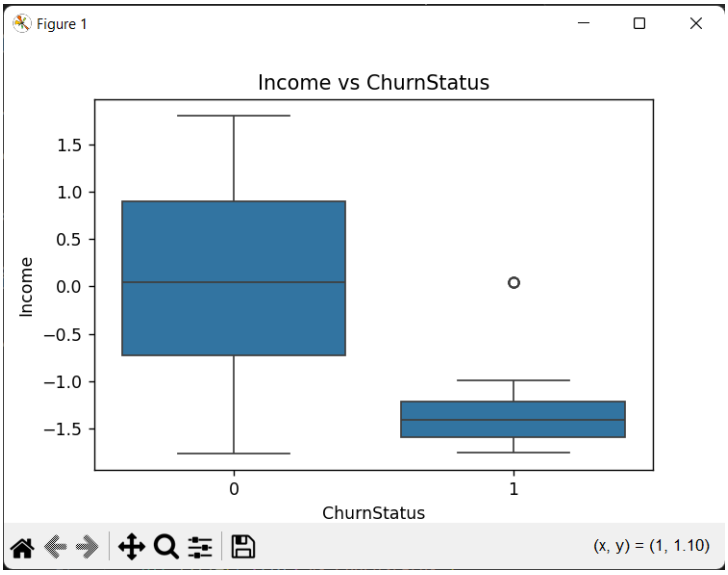


This is a histogram which shows the relationship between the (scaled) ChurnStatus and number as we show the most number is Stayes.

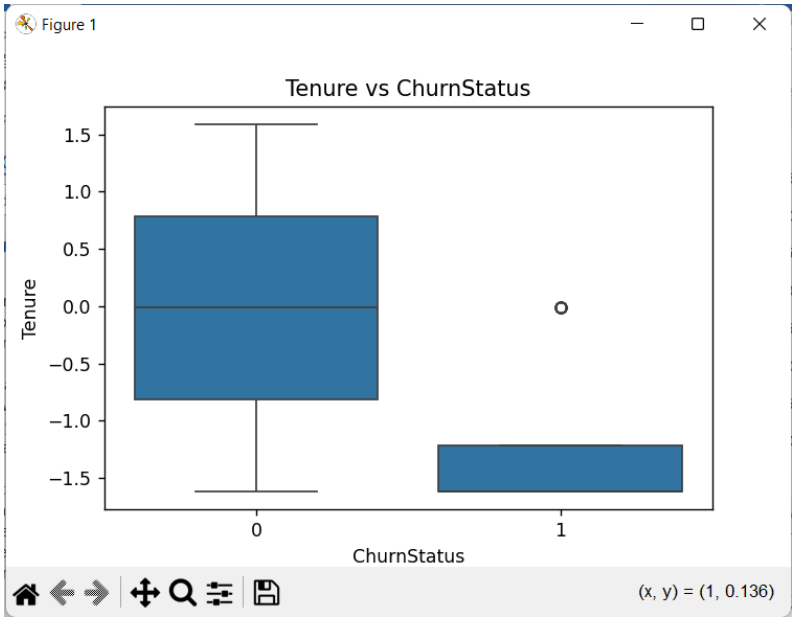
Bivariate Analysis



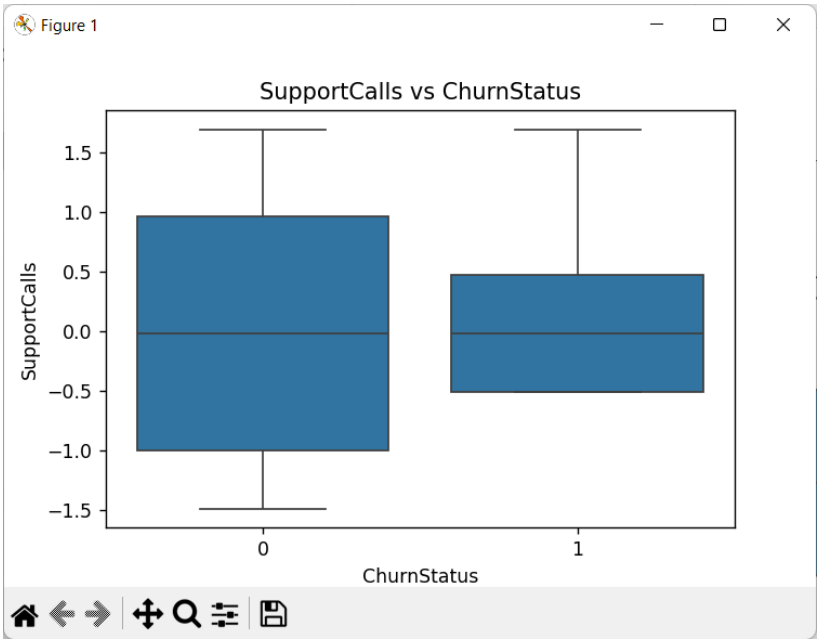
This is both boxes for churned and stayed customers there is no strong difference in age between customers who churn and those who stay.



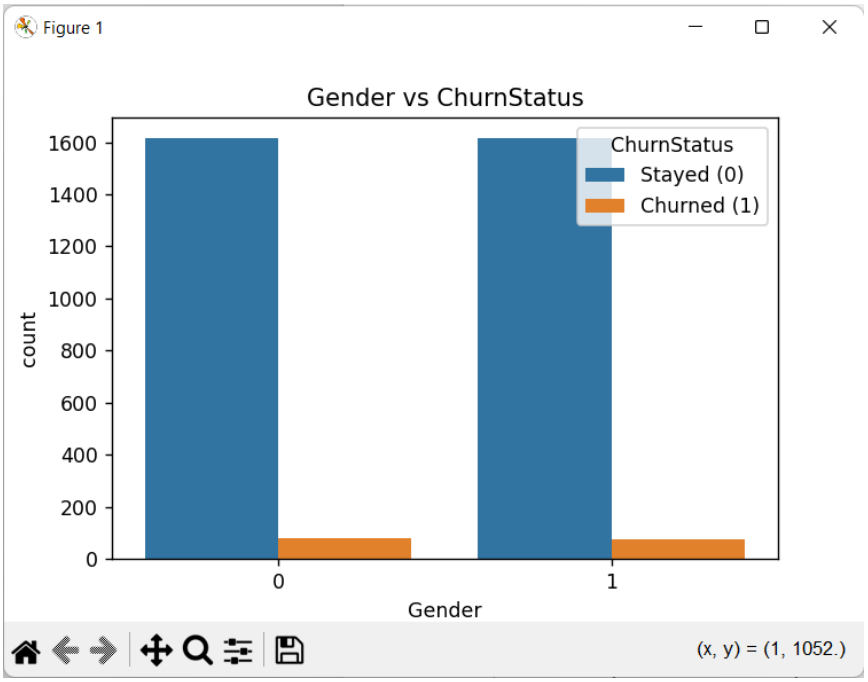
This is both boxes for churned and stayed customers with Income The median income for churned customers (ChurnStatus = 1) is lower than for those who stayed (ChurnStatus = 0) and there are one who has income equal 0(scaled) but he churned.



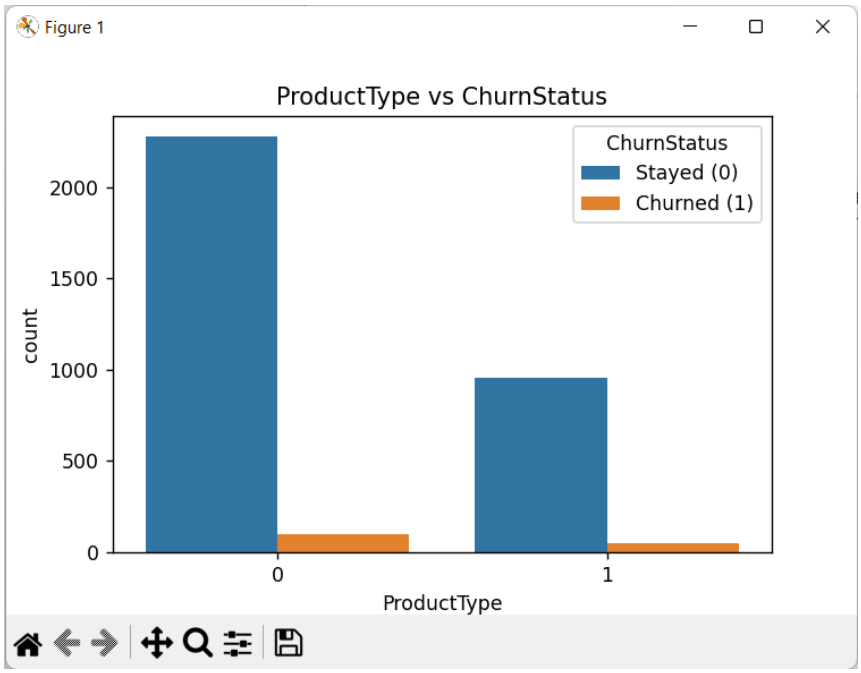
This is both boxes for churned and stayed customers with Tenure The difference between the two boxes is clear and significant and there are one who has Tenure equal 0(scaled) but he churned.



This is both boxes for churned and stayed customers with SupportCalls.

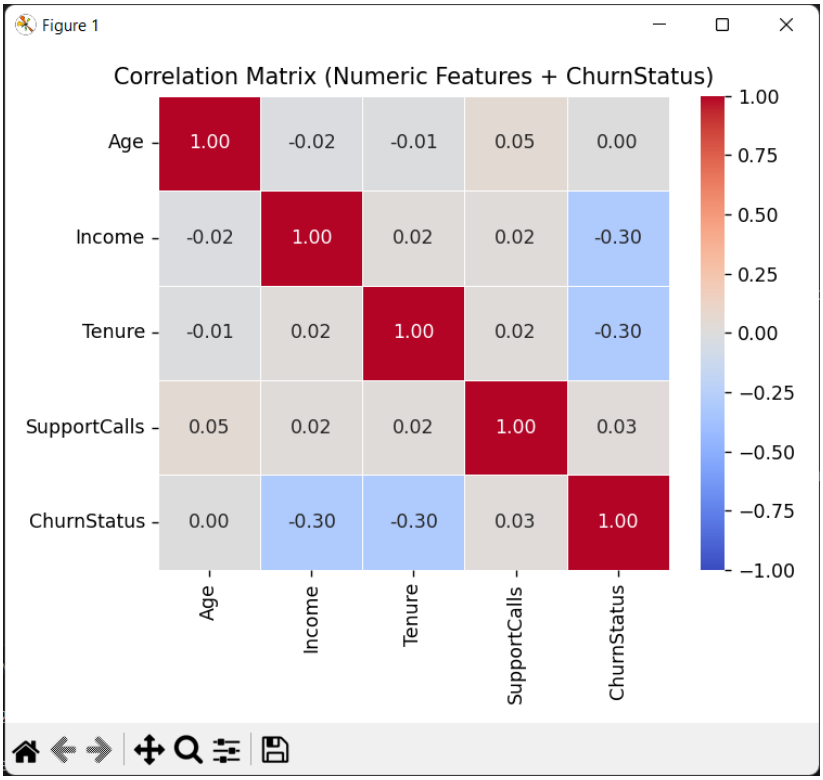


This is a Bar Plot to gender with Churnstatus There's no significant difference in churn behavior between genders.



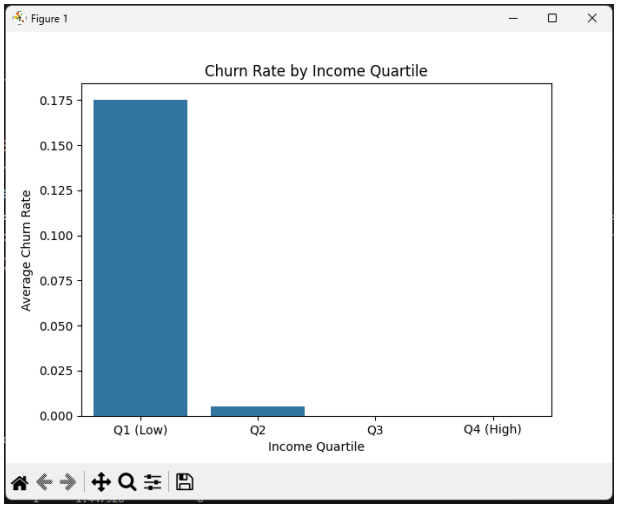
This is a Bar Plot to ProductType with Churnstatus as we show the most of product type 0 or 1 are stayed and little of them is chured.

Correlation Matrix

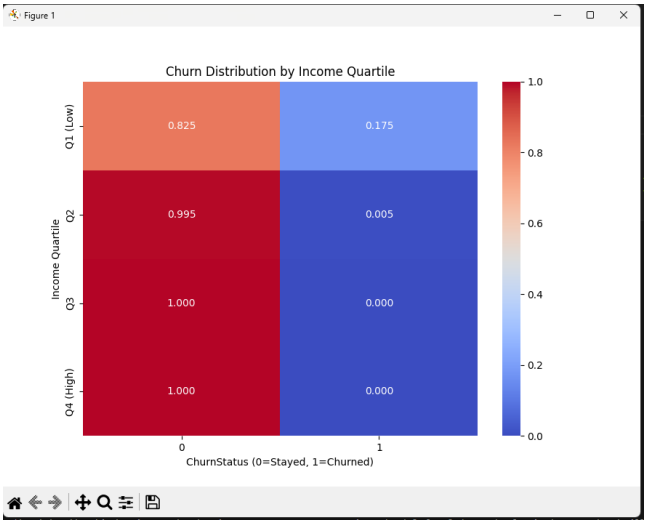


As we show in the Correlation Matrix the Income and the Tenure have the same and the most effect in churnStatus.

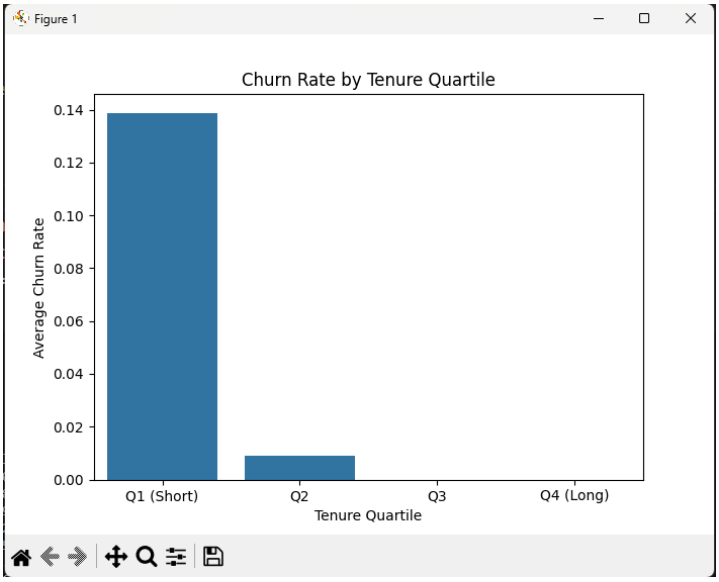
Task 6:



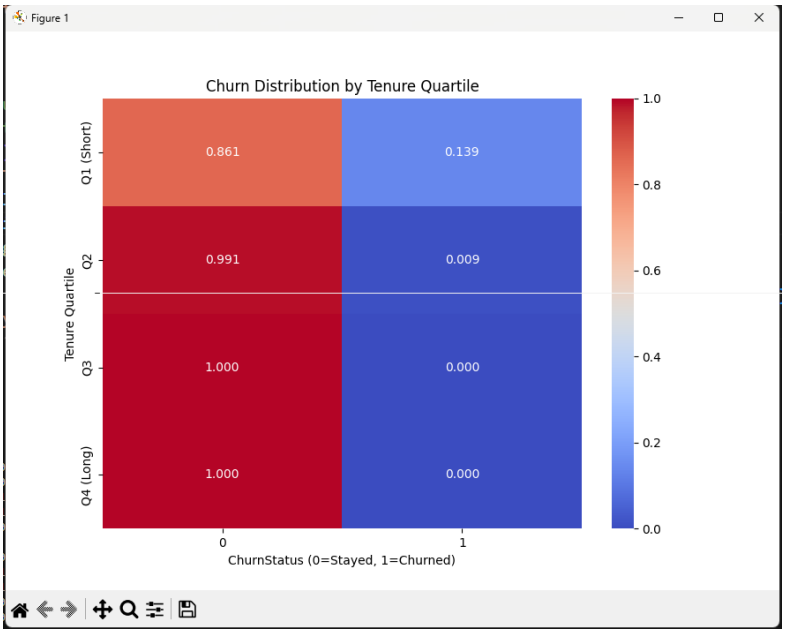
Lower-income customers are significantly more likely to leave the company, possibly due to cost sensitivity or limited perceived value.



Income has a strong negative relationship with churn — as income increases, customer retention improves.



Newer customers are more likely to churn, suggesting that improving early-stage customer experience could reduce churn.



Customer loyalty strengthens with time — retaining customers beyond their first tenure period greatly reduces churn risk.

Task 7:

As we show the Income and the Tenure have the most effect in churnStatus but we cannot change the income we can make offers, but to a certain extent. It may affect in churnStatus, but not significantly. we show in the both boxes for churned and stayed customers with Tenure the customers who is new is churned so We must take care of new customers so that we keep them and they will wait for us.